

```
===== Bank Client
=====
```

```
pki@pky:~/bank$ = openssl genpkey -algorithm RSA -out bank_pri.pem
```

```
C
pki@pky:~/bank$ = openssl rsa -pubout -in bank_pri.pem -out bank_pub.pem
-----
pki@pky:~/bank/acct1$ = openssl genpkey -algorithm RSA -out acct1_pri.pem
```

```
pki@pky:~/bank/acct1$ = openssl rsa -pubout -in acct1_pri.pem -out
acct1_pub.pem
```

```
-=====
```

```
pki@pky:~/bank/acct1$ = openssl pkeyutl -encrypt -pubin -inkey
../bank_pub.pem -in msg.txt -out msg.enc
```

```
mv msg.enc ../
```

```
pki@pky:~/bank$ = openssl pkeyutl -decrypt -inkey bank_pri.pem -in
msg.enc -out msg.txt
```

```
pki@pky:~/bank$ = ls
acct1 acct2 acct3 bank_pri.pem bank_pub.pem msg.enc msg.txt
```

```
===== RSA Sign and Verify
=====
```

```
pki@pky:~/bank$ = openssl genpkey -algorithm RSA -out pri.pem
```

```
pki@pky:~/bank$ = openssl rsa -pubout -in bank_pri.pem -out pub.pem
```

```
. openssl dgst -sha256 -sign keyfile.pem -out file1.txt.sha256.sig
file1.txt
```

```
. openssl dgst -sha256 -verify public_key.pem -signature
file1.txt.sha256.sig
file1.txt
```

```
===== Create certificates
rootca,sub,private =====
```

Root CA Certificate

- Generate root CA private key:
= openssl genpkey -algorithm RSA -out root.key
- Generate root CA Certificate Signing Request (CSR):
= openssl req -new -key root.key -out root.csr
- Self-sign the Root CA certificate:

```
= openssl x509 -req -in root.csr -signkey root.key -out root.crt -days 3650
```

```
· View the file in a readable format:  
.= openssl x509 -in root.crt -text -noout
```

Sub CA Certificate

```
· Generate sub CA private key:  
.= openssl genpkey -algorithm RSA -out sub.key  
  
· Generate sub CA Certificate Signing Request (CSR):  
.= openssl req -new -key sub.key -out sub.csr
```

```
· Sign the sub CA CSR with Root CA:  
.= openssl x509 -req -in sub.csr -CA root.crt -CAkey root.key -  
CAcreateserial -out sub.crt -days 1825
```

```
· View the file in a readable format:  
.= openssl x509 -in sub.crt -text -noout
```

Issue Digital Certificates from Sub CA

```
· Generate some server's private key:  
.= openssl genpkey -algorithm RSA -out server.key  
  
· Generate CSR for an end entity (e.g. some server): cdac.ac.in  
.= openssl req -new -key server.key -out server.csr  
  
· Sign the CSR with sub CA:  
· openssl x509 -req -in server.csr -CA sub.crt -CAkey sub.key -  
CAcreateserial -out server.crt -days 365
```

```
· View the file in a readable format:  
.= openssl x509 -in server.crt -text -noout
```

which apache2

```
---install apache2 -> restart > enable > status (systemctl -- apache2 )
```

edit the html file

```
vim /var/www/html/index.html
```

then creat ssl.conf file

```
cd /etc/apache2/site-available/
```

```
vim name.com-ssl.conf
```

```
<VirtualHost *:443>  
    ServerName uttu.com  
    ServerAlias www.uttu.com  
    DocumentRoot /var/www/html
```

```
        SSLEngine on
        SSLCertificateFile /home/ditiss/uttu/www.uttu.com.pem
        SSLCertificateKeyFile /home/ditiss/uttu/uttu_key.pem
        SSLCACertificateFile /home/ditiss/uttu/rootCA.crt
    </VirtualHost>
-----

a2enmod ssl
a2ensite name.com-ssl
systemctl restart apache2
systemctl enable apache2
systemctl reload apache2

---- go to browser and save the certificate and load the webiste
```